

Engineering demonstration and defense

Project: Grounded Learning Service

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone release review

Without opening the old lesson, explain one key idea from Capstone release review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Capstone service implementation

Without opening the old lesson, explain one key idea from Capstone service implementation. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Security and privacy review

Without opening the old lesson, explain one key idea from Security and privacy review. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace the worked case for Engineering demonstration and defense. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Engineering demonstration and defense. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Apply a new case independently

Defense records 8/10 case passes, no unauthorised calls in those traces and successful local rollback.
What can be claimed?

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. W1 Explain the mechanism

Explain the weekly mechanism in your own words. Use the worked scenario to name its input, decision rule and output.

My reasoning and evidence

6. W2 Trace the worked case

Demo succeeds on three cases using mocked storage. Presenter claims real network reliability and production security. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

7. W3 Audit the limit

A report omits this qualification: A defense earns credibility through traceable scope and honest failures, not universal guarantees. Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

8. W4 Apply a new case independently

Defense records 8/10 case passes, no unauthorized calls in those traces and successful local rollback. What can be claimed?

My reasoning and evidence

9. W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison. Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

10. Project checkpoint and reflection

For Grounded Learning Service, use this week's independent task as a rehearsal for the final evidence pack: Defense records8/10 case passes, no unauthorised calls in those traces and successful local rollback. What can be claimed? Keep actual observations separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

11. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

12. Pictorial case practice: Engineering demonstration and defense

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Approved card V7 says club V uses room E. All trusted tools are read-only.

Engineering demonstration and defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Show source path

Connect the question to retrieved card V7.



2. Support the claim

Room E follows from V7, not merely fluent wording.



3. Show authority

The service cannot edit rooms or send messages.



4. Name missing evidence

With no approved card, decline to invent a room.

CHANGE ONE THING

The challenger asks the bot to email everyone about room E.

Explain the changed result using the matching lesson.

Locate V7, room E and read-only scope in the demonstration. Which evidence supports the answer, and why must a request to email everyone still be refused?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

13. Project portfolio milestone

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with evidence.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or evidence that makes the deciding step valid, so another team can review it.

Evidence to keep · My record

Evidence to keep: This week's input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week's record rather than starting again.

14. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Defend one complete service trace. At the demonstration, a challenger asks how a room answer was produced and what the bot cannot do.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? With no approved card, decline to invent a room.

1. Name missing evidence
2. Show source path
3. Support the claim
4. Show authority

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: Name missing evidence With no approved card, decline to invent a room. A defence links behaviour to evidence and limitations. Demonstrate both a valid answer path and a meaningful refused request.

Practice 2: explain the deciding evidence

Changed-case practice: The challenger asks the bot to email everyone about room E. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future input.
2. Refuse that operation while retaining the supported room answer.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: Refuse that operation while retaining the supported room answer. Knowing a fact does not grant authority for an external action.

Repair a missing building block

Review Capstone release review (ai-6/35). Goal: Explain the mechanism and terminology of capstone release review. Short practice: Apply the three mandatory release gates to the recorded results. Why do two passes not outweigh the deletion failure, and what evidence must follow its repair?. Return to this week and explain what you changed.

Review Capstone service implementation (ai-6/33). Goal: Explain the mechanism and terminology of capstone service implementation. Short practice: Trace the G2 fact request and edit request through the prototype. What can be returned, what cannot be done, and how does removing G2 change those two outcomes?. Return to this week and explain what you changed.

Review Security and privacy review (ai-6/30). Goal: Explain the mechanism and terminology of security and privacy review. Short practice: Trace private note content and analytics counts to their permitted readers. Why is storing full notes in the dashboard outside this design, and what risk can remain for tiny-group counts?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
